

Chapter 5: Resampling Methods

Exercises 1 and 9

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Introduction

This report presents solutions to Exercises 1 and 9 from Chapter 5, *Resampling Methods*, of *An Introduction to Statistical Learning with Applications in Python (ISLP)*. Exercise 1 is a conceptual derivation involving the minimum-variance combination of two random variables. Exercise 9 applies bootstrap methods to the Boston housing data set to estimate the mean, median, and tenth percentile of the `medv` variable.

The calculations in Exercise 9 were implemented in Python using the ISLP package. The bootstrap calculations use 10,000 bootstrap samples, with the random number generator initialized using seed 1.

1 Exercise 1

Question

Using the basic statistical properties of the variance, as well as single-variable calculus, derive equation (5.6). In other words, show that the value of α given by equation (5.6) minimizes

$$\text{Var}(\alpha X + (1 - \alpha)Y).$$

Solution

Let

$$V(\alpha) = \text{Var}(\alpha X + (1 - \alpha)Y).$$

Define

$$\sigma_X^2 = \text{Var}(X), \quad \sigma_Y^2 = \text{Var}(Y),$$

and

$$\sigma_{XY} = \text{Cov}(X, Y).$$

Using the variance formula

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y),$$

we obtain

$$V(\alpha) = \alpha^2 \sigma_X^2 + (1 - \alpha)^2 \sigma_Y^2 + 2\alpha(1 - \alpha)\sigma_{XY}.$$

Expanding the terms gives

$$V(\alpha) = \alpha^2\sigma_X^2 + (1 - 2\alpha + \alpha^2)\sigma_Y^2 + (2\alpha - 2\alpha^2)\sigma_{XY}.$$

Collecting terms according to powers of α ,

$$V(\alpha) = \alpha^2(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) + \alpha(-2\sigma_Y^2 + 2\sigma_{XY}) + \sigma_Y^2.$$

To find the value of α that minimizes the variance, differentiate with respect to α :

$$\frac{dV}{d\alpha} = 2\alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) - 2\sigma_Y^2 + 2\sigma_{XY}.$$

At the minimum,

$$\frac{dV}{d\alpha} = 0.$$

Therefore,

$$2\alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) - 2\sigma_Y^2 + 2\sigma_{XY} = 0.$$

Dividing by 2,

$$\alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}.$$

Hence,

$$\hat{\alpha} = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}$$

which is the required result.

Verification that the solution is a minimum

We take the second derivative:

$$\frac{d^2V}{d\alpha^2} = 2(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}).$$

Notice that

$$\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY} = \text{Var}(X - Y).$$

Since variance is non-negative,

$$\text{Var}(X - Y) \geq 0.$$

Therefore,

$$\frac{d^2V}{d\alpha^2} \geq 0.$$

Thus, the value of $\hat{\alpha}$ obtained above minimizes $\text{Var}(\alpha X + (1 - \alpha)Y)$.

2 Exercise 9: Boston Housing Data

Data and Method

The Boston housing data set contains 506 observations. The variable `medv` represents the median value of owner-occupied homes for Boston census tracts, measured in thousands of dollars.

For parts (c), (f), and (h), bootstrap sampling was used. Each bootstrap sample contains 506 observations selected with replacement from the original data. The relevant statistic was calculated for each of 10,000 bootstrap samples. The bootstrap standard error was then estimated as the standard deviation of the resulting bootstrap statistics.

Python setup

The data were loaded as follows:

```
from ISLP import load_data
import numpy as np

Boston = load_data("Boston")
Boston.head()
```

(a) Estimate the population mean

The population mean is estimated using the sample mean:

$$\hat{\mu} = \bar{x}.$$

In Python:

```
mu_hat = Boston['medv'].mean()
print(mu_hat)
```

The result is

$$\hat{\mu} = 22.5328063.$$

Therefore, the estimated population mean of `medv` is approximately 22.53 thousand dollars.

(b) Estimate the standard error of $\hat{\mu}$

The standard error of the sample mean is estimated using

$$SE(\hat{\mu}) = \frac{s}{\sqrt{n}},$$

where s is the sample standard deviation and n is the number of observations.

The Python implementation is:

```
se_mu = Boston['medv'].std() / np.sqrt(Boston.shape[0])
print(se_mu)
```

The result is

$$SE(\hat{\mu}) = 0.4088611.$$

This means that the sampling variability of the estimated mean is about 0.409 thousand dollars.

(c) Bootstrap estimate of the standard error

The bootstrap procedure was used to estimate the standard error of the mean. The procedure repeatedly draws samples of size 506 with replacement and calculates the mean of each sample.

```

rng = np.random.default_rng(1)

bootstrap_means = []

for i in range(10000):
    sample = rng.choice(
        Boston['medv'],
        size=len(Boston),
        replace=True
    )

    bootstrap_means.append(sample.mean())

bootstrap_se = np.std(bootstrap_means, ddof=1)

print(bootstrap_se)

```

The bootstrap standard error is

$$SE_{\text{bootstrap}}(\hat{\mu}) = 0.4097871.$$

The analytical standard error from part (b) was 0.4088611, while the bootstrap estimate is 0.4097871. The two estimates are extremely close. This indicates that the bootstrap provides a good estimate of the sampling variability of the sample mean in this data set.

(d) 95% confidence interval for the population mean

The exercise suggests the approximate confidence interval

$$[\hat{\mu} - 2SE(\hat{\mu}), \hat{\mu} + 2SE(\hat{\mu})].$$

Using the bootstrap standard error from part (c),

$$\hat{\mu} = 22.5328063$$

and

$$SE_{\text{bootstrap}} = 0.4097871.$$

Therefore,

$$22.5328063 - 2(0.4097871) = 21.7132322$$

and

$$22.5328063 + 2(0.4097871) = 23.3523805.$$

Thus, the approximate 95% confidence interval is

$$[21.7132, 23.3524].$$

The Python calculation is:

```
lower = mu_hat - 2 * bootstrap_se
upper = mu_hat + 2 * bootstrap_se

print(lower, upper)
```

Hence, we estimate that the population mean of `medv` lies approximately between 21.71 and 23.35 thousand dollars.

(e) Estimate the population median

The population median is estimated using the sample median:

```
mu_med = Boston['medv'].median()
print(mu_med)
```

The result is

$$\hat{\mu}_{med} = 21.2.$$

Therefore, the estimated population median of `medv` is 21.2 thousand dollars.

(f) Bootstrap standard error of the median

Unlike the sample mean, there is no simple formula of the same form for the standard error of the median. Therefore, the bootstrap is used.

```
bootstrap_medians = []

for i in range(10000):
    sample = rng.choice(
        Boston['medv'],
        size=len(Boston),
        replace=True
    )

    bootstrap_medians.append(np.median(sample))

bootstrap_median_se = np.std(
    bootstrap_medians,
    ddof=1
)

print(bootstrap_median_se)
```

The bootstrap estimate of the standard error is

$$SE_{\text{bootstrap}}(\hat{\mu}_{med}) = 0.3813021.$$

This shows that the estimated median has sampling variability of about 0.381 thousand dollars. The bootstrap is useful here because it provides a practical estimate of uncertainty without requiring a simple analytical formula for the median's standard error.

(g) Estimate the tenth percentile

The tenth percentile is estimated using NumPy's `percentile()` function:

```
np.percentile(Boston['medv'], 10)
```

The result is

$$\hat{\mu}_{0.1} = 12.75.$$

Thus, approximately 10% of the observations have a `medv` value below 12.75 thousand dollars.

(h) Bootstrap standard error of the tenth percentile

The bootstrap was again used, this time calculating the tenth percentile for every bootstrap sample.

```
bootstrap_percentiles = []

for i in range(10000):
    sample = rng.choice(
        Boston['medv'],
        size=len(Boston),
        replace=True
    )

    bootstrap_percentiles.append(
        np.percentile(sample, 10)
    )

bootstrap_percentile_se = np.std(
    bootstrap_percentiles,
    ddof=1
)

print(bootstrap_percentile_se)
```

The resulting bootstrap standard error is

$$SE_{\text{bootstrap}}(\hat{\mu}_{0.1}) = 0.4999335.$$

This indicates that the estimated tenth percentile has a sampling variability of approximately 0.500 thousand dollars. The standard error is larger than the standard error of the mean because percentile estimation, especially toward the lower tail of a distribution, can be more variable.

3 Summary of Results

Part	Quantity	Result
9(a)	Sample estimate of population mean	22.5328
9(b)	Analytical standard error of mean	0.4089
9(c)	Bootstrap standard error of mean	0.4098
9(d)	Approximate 95% confidence interval	[21.7132, 23.3524]
9(e)	Sample estimate of population median	21.2
9(f)	Bootstrap standard error of median	0.3813
9(g)	Tenth percentile	12.75
9(h)	Bootstrap standard error of tenth percentile	0.4999

Table 1: Results obtained from the Boston housing data.

4 Conclusion

Exercise 1 demonstrates how basic variance properties and differentiation can be used to obtain the minimum-variance weight for a combination of two random variables. The resulting estimator is

$$\hat{\alpha} = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Exercise 9 demonstrates the practical value of the bootstrap. For the Boston housing data, the bootstrap estimate of the standard error of the mean (0.4098) was very close to the analytical standard error (0.4089). The bootstrap was also used to estimate the standard errors of the median and tenth percentile, for which a simple standard-error formula is not directly available in the exercise.

Overall, the results show how resampling can be used to quantify the uncertainty associated with statistical estimates using repeated samples drawn from the observed data.

Reference

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. *An Introduction to Statistical Learning with Applications in Python*. Springer, 2023.